THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 1: **Exploration: Entering the World of Chemistry** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is a derived unit? 1

(a) m/s **(b)** kg **(c)** s **(d)** m

2. A student estimates the height of a building as 30 m by comparing with a 3 m floor and counting 10 floors. This is: 1

(a) a law **(b)** an estimate by comparison

(c) an exact measurement **(d)** a theory

3. The least count of a ruler marked in mm is: 1

(a) 1 mm **(b)** 10 mm **(c)** 0.1 mm **(d)** 1 cm

4. The prefix 'milli' means: 1

(a) one-hundredth **(b)** one-thousandth **(c)** million **(d)** thousand

5. The number of seconds in a leap year is: 1

(a) 3153600 **(b)** 31536000 **(c)** 86400 **(d)** 31622400

6. The thickness of a sheet is found by measuring a stack of 100 sheets as 1.2 cm. The thickness of one sheet is: 1

(a) 0.12 cm **(b)** 1.2 cm **(c)** 0.0012 cm **(d)** 0.012 cm

7. The prefix 'micro' means: 1

(a) one-thousandth **(b)** one-hundredth **(c)** one-millionth **(d)** million

8. The mass of 3 litres of water (1 kg per litre) in grams is: 1

(a) 300 **(b)** 30000 **(c)** 30 **(d)** 3000

9. Which of the following is an assumption in a model of a falling ball? 1

(a) the ball is a gas **(b)** the ball has no mass

(c) air resistance is negligible **(d)** the ball is magnetic

10. The number of micrometres in a metre is: 1

(a) 1000 **(b)** 10^9 **(c)** 100 **(d)** 1000000

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why must an experiment be repeated by different scientists? 2

12. Why is the Sun's mass estimated rather than measured directly? 2

13. State two limitations of the scientific method. 2

14. Estimate the number of heartbeats in a lifetime of 70 years, taking 72 beats per minute (approximately). 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why scientists use models like the solar system for atoms and why such models have limits. 3

16. Explain the relationship between observation, hypothesis, prediction, experiment and conclusion using an example of rainfall. 3

17. Explain with an example how an estimate can be used to check whether a calculated answer is reasonable. 3

OR

Convert: 3
(a) 108 km/h to m/s
(b) 15 m/s to km/h
(c) 5 g/cm3 to kg/m3.

18. State three reasons why science models of weather become less accurate for long forecasts. 3

OR

State why science questions are tested using evidence and not authority alone. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. A student measures the thickness of a coin as 1.82, 1.84 and 1.83 mm. Calculate the mean, the percentage error from the accepted 1.80 mm and discuss precision and accuracy. 5

OR

Explain with a numerical example the use of significant figures and rounding in measurements. 5

20. Explain how mathematics, observation and experiment together led to the discovery of one law, such as Newton's gravitation or Boyle's law. 5

OR

Explain how chemistry as a science grows through questions, experiments, models and applications; mention two modern applications. 5

END OF QUESTION PAPER — Total: 40 Marks

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